



One in every six

What to do

Manufacturers of a chocolate bar claim that one bar in every six is a winner and contains a special imprint on the inside of the wrapper. Can you devise an experiment to test this claim?

- 1 What is the (theoretical) probability of purchasing a chocolate bar with a special wrapper?
- 2 Your friend carried out 100 trial purchases of chocolate bars by rolling a fair 6-sided die. Obtaining a 6 represented a winning chocolate bar, and numbers 1–5 represented a non-winning chocolate bar.

The results are shown in the table below.

Outcome	Frequency
Winning chocolate	16
Non-winning chocolate	84
Total	100

Calculate the relative frequency of the winning chocolate.

- 3 Compare the results obtained in parts 1 and 2.
- 4 Repeat the experiment conducted by your friend in part 2.
- 5 How do the results you obtained compare with the theoretical value and your friend's results?
- 6 What other devices could be used to conduct this experiment? Explain what must be taken into consideration when selecting possible devices?

Discuss your thinking

Would you expect the results to vary if 200 trials or 1000 trials were carried out? If so, why?

You will need

- A 6-sided die
- A scientific or graphical calculator